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Establishment of a Time-Varying Modified Kanai-Tajimi Non Stationary Stochastic Model of Seismic Records Based on the S-transform

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This article investigates the time-varying characteristics of seismic records. Time-varying amplitude and energy spectrums are defined to reflect the time-frequency dependence, and a general formulation of the S-transform is introduced. The S-transform is tested with various window functions to analyse the Kobe seismic records. The results indicate that using a complex window function with properly adjusted parameters gives favorable outcomes. Analyses of three soil sites show that sites with hard soil feature seismic records with shorter stationary durations, higher frequency centers, and broader frequency bands than other soil sites. The average time-varying spectrums of the seismic records are simulated using a uniform non stationary stochastic model and a time-varying modified non stationary Kanai-Tajimi stochastic model. Empirical formulas are established for the time-varying spectrums of the earthquake records from these sites by nonlinearly fitting stochastic models to the record data. The values of the time-varying spectrum factors for different earthquake intensities and sites agreeing with the Chinese Seismic Code are obtained. Based on these analyses and observations, we propose using the solutions to the stochastic models to simulate non stationary ground motions.

Keywords S-transform; Non Stationary Stochastic Model; Time-Varying Spectrum; Seismic Record

1. Introduction

Due to the dissipation of seismic waves along their propagation paths and the difference of the arrival times of P-waves, S-waves and surface waves, seismic records are non stationary in both amplitude and frequency. This influences the seismic responses of linear and nonlinear structures [Yeh and Wen, 1990; Wang *et al.*, 2002]. Traditional analysis methods based on the Fourier transform are incapable of capturing this time-frequency locality of non stationary signals. Researchers employ time-frequency analysis tools to study the time-varying energy spectrums of seismic records. The most widely used time-frequency analysis tools include the wavelet transform, Hilbert-Huang transform (HHT), S-transform, and adaptive Chirplet transform. Iyama and Kuwamura [1999] and Mukherjee and Gupta [2002] determined the relationship between the time-varying energy spectrum and the continuous wavelet coefficients of seismic records using the orthogonal wavelet basis. To overcome the wavelet transform's redundancy and the algorithm's slowness,

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Gurley and Kareem [1999] determined the relationship between the mean squares of the discrete wavelet coefficients and the time-varying energy spectrum. Spanos *et al.* [2005] and Spanos and Failla [2004] estimated the time-varying spectrums of seismic records using a binary harmonic wavelet transform, an ordinary harmonic wavelet transform, and a filtering harmonic wavelet transform and identified the mathematical relationship between the time-varying energy spectrum and the wavelet coefficients. Recently, these authors also studied the time-frequency characteristics of seismic records and the seismic responses of structures using the adaptive Chirplet transform [Spanos *et al.*, 2007]. Wen and Gu [2004] decomposed non stationary seismic records using the HHT and expressed the time-frequency locality by defining the Hilbert amplitude spectrum, the Hilbert energy spectrum, and the Hilbert marginal spectrum. Suarez and Montejó [2005] developed a wavelet procedure to generate an accelerogram whose response spectrum is compatible with a target spectrum. Neild *et al.* [2003] reviewed analytical tools of the time-frequency distribution for nonlinear vibrations of structures. Rezaeian and Der Kiureghian [2008] developed a fully non stationary stochastic model to describe earthquake ground motions. The model is based on the modulation of the response of a linear filter with time-varying characteristics to a discretized white-noise excitation. The model parameters are estimated by fitting selected statistical characteristics of a target accelerogram. This model is a stochastic ground motion model based on time domain, and it couldn't be used in the frequency domain analysis of random earthquake response of structure.

Some researchers directly have deduced non separable stochastic seismic model and separable stochastic seismic model by using the relationship between the proposed stochastic model and the response spectrum of the seismic code in recent years. Cacciola and Zentner [2012] chose the evolutionary Clough-Penzien model as non separable stochastic seismic model, the time-varying characteristics frequency of site soil was expressed by using the exponential function presented by Ahmadi and Fan [1990], and the parameters of the model are random variable calibrated on a set of recorded earthquake; Giaralis and Spanos [2009, 2012] and Spanos *et al.* [2009] took the Kanai-Tajimi spectrum, Clough-Penzien spectrum, and Butterworth filtered spectrum modulated by BGB envelope function as separable stochastic seismic model, and the elastic displacement response spectrum in EC8 and Chinese Seismic Code is treated as the targeted spectrum, the parameters of the model were calibrated by solving a least squares minimization problem, which the objective function was the minimized square of the difference between the targeted spectrum and the maximum displacement response of structure. The non stationary stochastic seismic model can be obtained from the above models by using the response spectrum of seismic code, however, only the frequency characteristics of seismic motion is included in the response spectrum of seismic code, which can't describe the time-frequency non stationary characteristics of earthquake ground motion. Consequently, the precision of the seismic model based on the above methods is depended on the reliability of the selected time-varying functions (such as the time modulating function, the time-varying characteristics frequency, and the damping ratio of site soil). The reliable time-varying functions can be obtained through the time-frequency analysis and statistical analysis for lots of seismic records, which come from different sites characteristics, moment magnitude of the earthquake, and the closest distance from the recording site to the ruptured area.

The S-transform is a type time-frequency analysis tool that has been developed in recent years. It adopts the variable-scale localized Gaussian function to replace those used in Stockwell *et al.* [1996], Stockwell [2007], Pinnegar [2005], and Pinnegar and Masintha [2004].

As a result, the time-frequency resolution of the S-transform changes with the frequency. In spite of its popularity in many other fields, the S-transform has not been applied

to investigate the time-frequency characteristics of seismic records. In this article, the basic algorithm of the S-transform is introduced, and the time-varying amplitude spectrum and the time-varying energy spectrum are defined to reflect the time-frequency characteristics of the seismic record. The time-frequency characteristics of the Kobe seismic record are then analyzed based on the S-transform with different window functions. Based on the S-transform, the time-varying spectrums of large numbers of seismic records are estimated. The average time-varying spectrums of the seismic records are simulated using the uniform non stationary stochastic model and the time-varying modified Kanai-Tajimi non stationary stochastic model [Fan and Ahmadi, 1990]. The empirical formulas for the time-varying spectrums are established for three classes of soil sites using a nonlinear fitting method. Finally, the artificial non stationary ground motion is generated by solving the time variable linear differential equations.

2. S-transform and Time-Varying Spectrum

If the seismic time history record is denoted as $s(t)$, the S-transform of the time history of the seismic records can be defined as:

$$S(f, \tau) = \int_{-\infty}^{+\infty} s(t)w(f, \tau - t, \sigma) \cdot \exp(-j2\pi ft)dt, \quad (1)$$

where f , t , and τ are the frequency, time, and center of the window function, respectively, and $w(f, \tau - t, \sigma)$ is the window function that can take different functional forms as below.

- a. The window width does not change with the frequency f :

$$w(\tau - t) = \frac{1}{\sqrt{2\pi}} \cdot \exp\left(\frac{-(\tau - t)^2}{2}\right); \quad (2)$$

in this case, Eq. (1) becomes the short-time Fourier transform (STFT).

- b. The window width changes with f , and there is no regulatory factor σ :

$$w(f, \tau - t) = \frac{|f|}{\sqrt{2\pi}} \exp\left(\frac{-f^2(\tau - t)^2}{2}\right); \quad (3)$$

in this case, Eq. (1) is the conventional definition of the S-transform.

- c. The window changes with f , and the change rate of its width can be adjusted by σ :

$$w(f, \tau - t, \sigma) = \frac{\sigma |f|}{\sqrt{2\pi}} \exp\left(\frac{-\sigma f^2(\tau - t)^2}{2}\right); \quad (4)$$

in this case, Eq. (1) is called the generalized S-transform.

- d. The window is a complex function whose width can be adjusted according to [16]:

$$w(f, \tau - t, \sigma) = \begin{cases} p \cdot \exp\left[-\frac{f^2(\tau - t)^2}{2}\right] \cdot \exp\{-j2\pi f(\tau - t) + j2\pi \cdot \text{sign}(f) \cdot \sigma \log[\sigma + |f|(\tau - t)]\}, & t < \tau + \sigma/|f| \\ 0 & t \geq \tau + \sigma/|f| \end{cases} \quad (5)$$

where p is a constant, and σ is an adjustable parameter that controls the size of the window. In this case, Eq. (1) is called the complex window generalized S-transform.

If the window function satisfies the condition:

$$\int_{-\infty}^{+\infty} R\{w(f, \tau, \sigma)\} d\tau = 1 \quad (6)$$

where $R\{\cdot\}$ extracts the real part, then $S(f, \tau)$ has the following property:

$$F(f) = \int_{-\infty}^{+\infty} S(f, \tau) \cdot d\tau, \quad (7)$$

where $F(f) = \int_{-\infty}^{+\infty} s(t) \cdot \exp(-j2\pi ft) dt$, which is the Fourier transform of $s(t)$.

Using Eq. (7), the inverse S-transform based on the inverse Fourier transform can be written as:

$$s(t) = \int_{-\infty}^{+\infty} \left(\int_{-\infty}^{+\infty} S(f, \tau) d\tau \right) \cdot \exp(j2\pi ft) df \quad (8)$$

In numerical calculations, Eqs. (1), (7), and (8) are usually converted into the discrete forms:

$$S[f, \tau] = \sum_{t=0}^{N-1} s[t] \cdot w[f, \tau - t, \sigma] \cdot \exp\left(\frac{-j2\pi ft}{N}\right) \quad (9)$$

$$F[f] = \sum_{\tau=0}^{N-1} S[f, \tau] \quad (10)$$

$$s[t] = \frac{1}{N} \sum_{f=-N/2}^{N/2-1} \cdot \sum_{\tau=0}^{N-1} S[f, \tau] \cdot \exp\left(\frac{j2\pi ft}{N}\right), \quad (11)$$

where N is the sampling number of the signal.

For easy calculation, the discrete S-transform can be decomposed into the product of two parts:

$$S[f, \tau] = A[f, \tau] \otimes e^{i\theta[f, \tau]}, \quad (12)$$

where $A[f, \tau] = |S[f, \tau]|$ is called the time-varying amplitude spectrum, $\theta[f, \tau]$ is called the time-varying phase spectrum, and $\otimes$ is the multiplication sign of the matrix.

Similar to the definition of a spectrogram based on the STFT, the time-varying energy spectrum based on the S-transform can be defined as:

$$P[f, \tau] = |S[f, \tau]|^2. \quad (13)$$

To exhibit the power of the S-transform-based time-frequency analysis methods in identifying the temporal features in the energy spectrums of seismic records, this paper applies these methods to the Kobe seismic record, which has a sampling period of 0.02 s, a sample size of 1000, and a peak acceleration value that is adjusted to 1 m/s².

The time history and the power spectrum density (PSD) are shown in Figs. 1a and 1b. As seen in Fig. 1b, the maximum peak value of the PSD occurs at 1.465 Hz, and the second highest peak value occurs at a higher frequency of 2.88 Hz. However, it is not possible to locate these positions in the time domain from either the time history record or the power spectrum when these frequency components occur in a PSD.

The S-transform is tested with different windows to produce the time-varying energy spectrum of the Kobe seismic record (Figs. 1c–f). Figure 1f shows the time-varying spectrum obtained by the complex window generalized S-transform, where σ is assumed to be 4. Figure 1f features four peaks in the time-frequency domain, where the highest peak has a frequency of 1.46 Hz and occurs at approximately 4.5 s, the second highest peak value has a frequency of 2.8 Hz and occurs at approximately 6.3 s, the third highest peak has a frequency of 1.6 Hz and occurs at approximately 7.8 s, and the fourth highest peak has a frequency of 2.25 Hz and occurs at approximately 3.8 s. Figure 1d shows the time-varying spectrum obtained by the S-transform. The maximum peak value at the time-frequency point (1.46 Hz, 4.5 s) is not as clear as that indicated by the complex window generalized S-transform. The peak value at the time-frequency point (1.6 Hz, 7.8 s) cannot be resolved clearly by the generalized S transform, as shown in Fig. 1e. The STFT (Fig 1c) can resolve the maximum peak value at the time-frequency point (4.5 s, 1.46 Hz) but cannot resolve the peak value at the time-frequency points (7.8 s, 1.6 Hz) and (6.3 s, 2.8 Hz). These observations reveal that the complex window generalized S-transform provides better time-frequency resolution than the S-transform and the STFT; for this reason, the complex window generalized S-transform is used in the following analysis.

To justify the time-frequency localization capabilities of complex window generalized S-transform for the representation of earthquake accelerograms, Fig. 2 shows the time-varying spectrums of Kobe earthquake record obtained by the dyadic harmonic wavelet transform (DHWT) [Spanos *et al.*, 2005; Newland, 1994], the generalized harmonic wavelet transform (GHWT) [Spanos *et al.*, 2005] and the adaptive optimal kernel

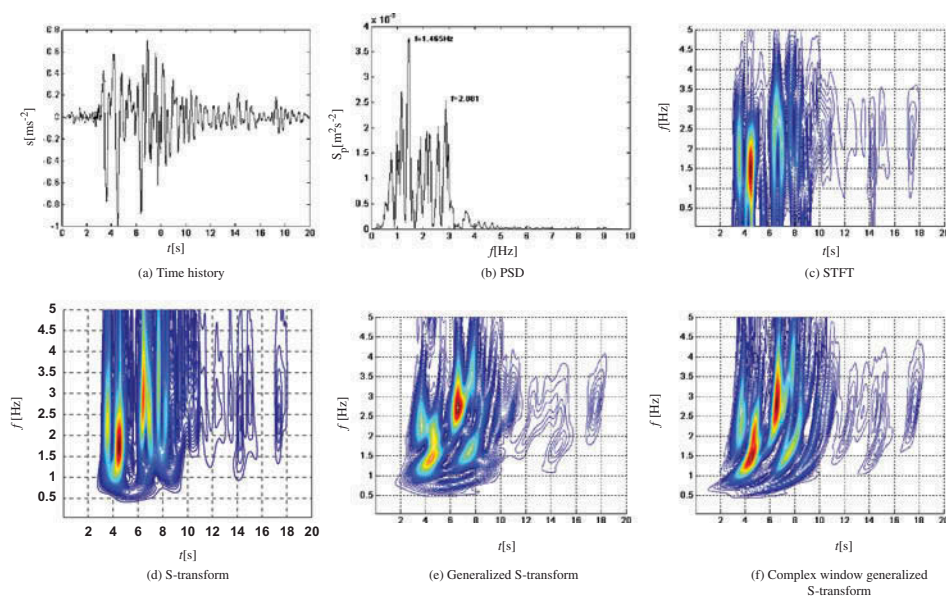


FIGURE 1 The time history, power spectrum, and time-varying spectrums of the Kobe earthquake record with different window functions.

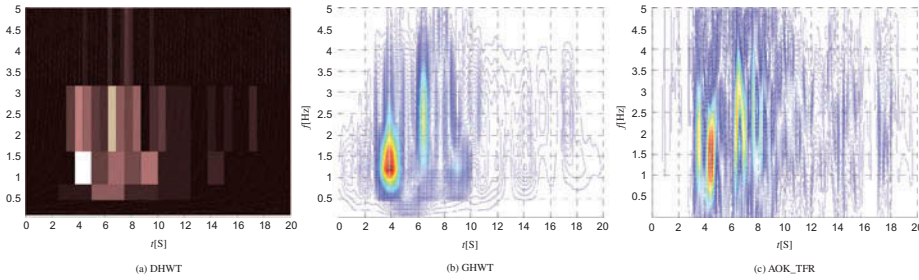


FIGURE 2 Time-varying spectra of the Kobe earthquake record obtained by DHWT, GHWT and AOK_TFR, respectively.

time-frequency representation (AOK_TFR) [Jones and Baraniuk, 1995]. This figure shows that DHWT can distinguish the approximate location of each time-frequency peak and has low time-frequency resolution; GHWT can just only distinguish the location of the maximum and the second peak accurately; AOK_TFR has the same time-frequency localization capabilities with complex window generalized S-transform and the locations of each peak are in accordance with that in Fig. 1f, which indicates that complex window generalized S-transform has high reliability.

3. Characteristics of the Time-Varying Spectrum

The time-varying spectra of large numbers of actual seismic records are generated using Eq. (13) and undergo a thorough statistical examination. The seismic records are divided into three classes as shown in Table 1. All of the seismic records are selected from the PEER Strong Motion Database (<http://peer.berkeley.edu/smcat/search.html>). Class 1 consists of seismic records from hard soil sites, class 2 consists of records from medium hard sites, and class 3 consists of records from soft soil sites. These records are standardized to have a unit energy. The standardized time history of a seismic record $\bar{x}(t)$ can be written as:

$$\bar{x}(t) = x(t) / \sqrt{\int_0^{t_p} x^2(t) dt}, \quad (14)$$

where $x(t)$ is the original time history of the seismic record, and t_p is the duration (40 s in this article).

The mean of the time-varying spectrum $E[P(f, t)]$ is calculated for the standardized seismic records on different class sites (Figs. 3a–c). The figure also shows the time marginal function $\bar{X}^2(t)$ and the frequency marginal function $\bar{F}^2(f)$, which are defined as follows:

TABLE 1 Classification of seismic records

Site classification	Average shear wave velocity of soil $V_{sm} (m/s^{-1})$	Magnitude (M_w)	Epicentral distance (km)	Number
Class 1	$V_{sm} > 500$	6~7	20~50	81
Class 2	$150 \leq V_{sm} \leq 500$	6~7	20~50	125
Class 3	$V_{sm} < 150$	6~7	20~50	65

$$\bar{X}^2(t) = \int_0^{+\infty} E[P(f, t)] df \quad (15)$$

$$\bar{F}^2(f) = \int_0^{t_p} E[P(f, t)] dt. \quad (16)$$

The time marginal function, the frequency marginal function, and the average time-varying spectrum reflect the energy distribution of the seismic records in the time domain, frequency domain, and time-frequency domain, respectively. As seen in Fig. 3, the records from the harder sites have a longer stationary duration in the time domain and a higher frequency center and broader frequency band in the frequency domain.

To study the non stationary characteristics of the seismic records, Fig. 4 shows the instantaneous center frequency $f_i(t)$ and the instantaneous bandwidth $2\Delta f(t)$ of the seismic records. The functions $f_i(t)$ and $\Delta f(t)$ are defined as:

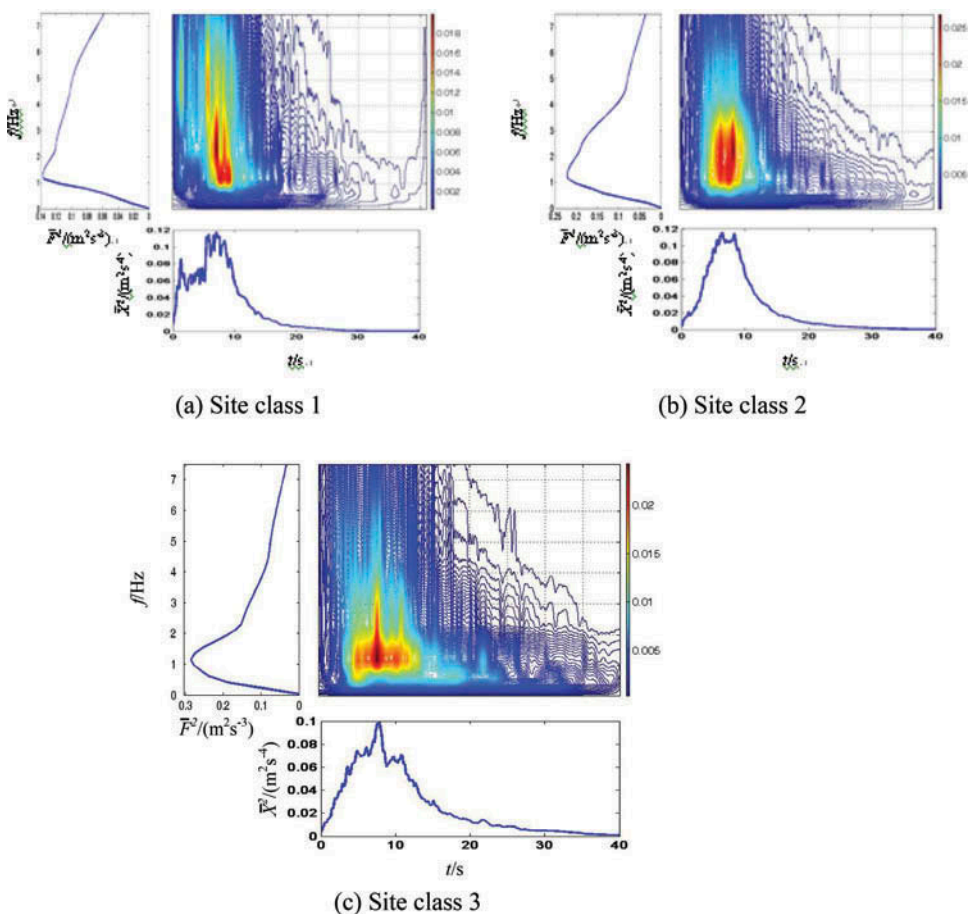


FIGURE 3 Time marginal function, frequency marginal function, and time-varying spectrum of actual seismic records.

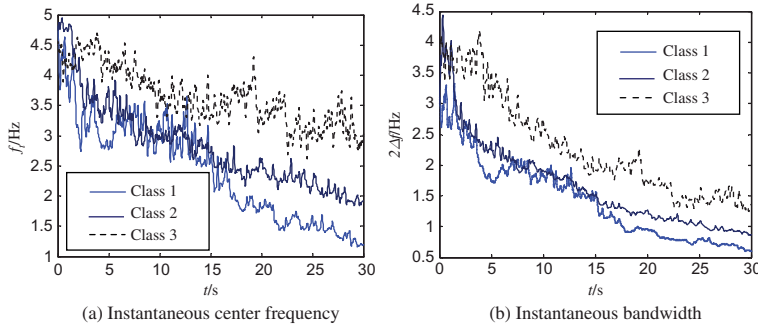


FIGURE 4 Instantaneous center frequency and instantaneous bandwidth of actual seismic records.

$$f_i(t) = \frac{\int_0^{+\infty} f \cdot E[P(f, t)] df}{\int_0^{+\infty} E[P(f, t)] df} \quad (17)$$

$$\Delta f^2(t) = \frac{\int_0^{+\infty} (f - f_i(t))^2 \cdot E[P(f, t)] df}{\int_0^{+\infty} E[P(f, t)] df}. \quad (18)$$

Figure 4 shows that $f_i(t)$ and $2\Delta f(t)$ decrease with time. This implies that the seismic records consist of higher frequency components in the initial stage and lower frequency components in the later stage. This observation also confirms that the seismic records have non stationary amplitudes and frequencies.

4. Construction of the Non Stationary Model

4.1. Uniform Non Stationary Stochastic Model of Standardized Seismic Records

The time-varying spectrum of the standardized seismic records is simulated using the uniform modulated non stationary model, which is formulated as:

$$S_p(f, t) = S_{k-T}(f, Z) \cdot g^2(t, C) \quad (19)$$

where $S_{k-T}(f, Z)$ is the modified Kanai-Tajimi model proposed by Clough and Penzien [1993], and $g(t, C)$ is the time modulating function:

$$S_{k-T}(f, Z) = \frac{f_g^4 + 4\zeta_g^2 f_g^2 f^2}{(f_g^2 - f^2)^2 + 4\zeta_g^2 f_g^2 f^2} \cdot \frac{f^4}{(f_f^2 - f^2)^2 + 4\zeta_f^2 f_f^2 f^2} S_{g0} \quad (20)$$

$$g(t, C) = \begin{cases} (t/t_1)^2 & 0 \leq t \leq t_1 \\ 1 & t_1 \leq t \leq t_2 \\ e^{-c(t-t_2)} & t \geq t_2 \end{cases}, \quad (21)$$

where $Z = (f_g, \xi_g, S_{g0})$ is the parameter set of $S_{k-T}(f, Z)$ and $C = (t_1, t_2, c)$ is the parameter set of $g(t, C)$. The parameter ξ_f is usually taken to be equal to ξ_g , while f_f is usually taken to be equal to $0.1 \sim 0.2 f_g$ [Cacciola and Zentner, 2012; Fan and Ahmadi, 1990] and is set at $0.15 f_g$ in this study.

In Eq. (19), the values of parameter set Z and parameter set C are determined for the seismic records from different sites. The solution process can be completed using the Matlab function lsqcurvef (·) for nonlinear curve fitting.

The time-varying spectrum parameter values of three types of records are identified, as shown in Table 2. Figure 5 shows the simulated time-varying spectrum, the time marginal function, and the frequency marginal function.

To quantify the accuracy of the simulated time-varying spectrum, the error function is defined as follows:

$$eps = \sum_{i=1}^n \sum_{j=1}^m (S_P(t_i, f_j) - E[P(t_i, f_j)])^2 \bigg/ \sum_{i=1}^n \sum_{j=1}^m (E[P(t_i, f_j)])^2. \tag{22}$$

TABLE 2 Time-varying spectral parameters of the standardized seismic records

Site classification	$f_g(\text{Hz})$	ζ_g	$S_{g0}(\text{m}^2\text{s}^{-3})$	$t_1(\text{s})$	$t_2(\text{s})$	C	Error (%)
Class 1	3.19	0.88	0.0108	5.06	7.91	0.12	8.2
Class 2	1.94	0.94	0.0161	4.60	8.51	0.11	4.3
Class 3	1.27	1.07	0.0163	3.38	9.50	0.07	6.7

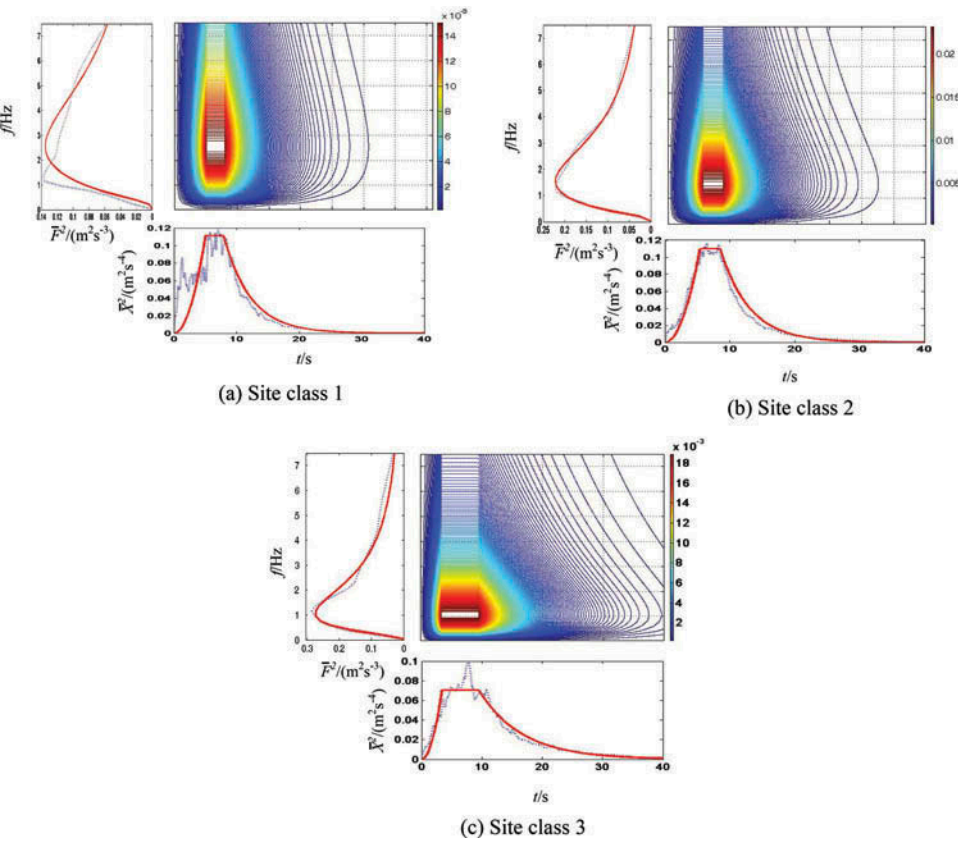


FIGURE 5 Time marginal function, frequency marginal function, and uniform non stationary stochastic model contour of seismic records.

The error values of the simulated time-varying spectrums on the three classes of sites are shown in Table 2. A comparison of Figs. 3 and 5 shows that the simulated time-varying spectrum agrees well with that of the actual seismic records.

4.2. Time-Varying Modified Kanai-Tajimi Model of Standardized Seismic Records

As explained in Sec. 3, both the amplitude and frequency of the seismic record are non stationary. In Eq. (19), the uniform non stationary stochastic model of the seismic record is stationary in the frequency domain, but the characteristic frequencies f_g and f_f and the characteristic damping ratios ξ_g and ξ_f do not change with time. In reality, however, the parameters f_g , f_f , ξ_g , and ξ_f are functions of time. Therefore, the time-varying modified Kanai-Tajimi model is proposed to simulate the time-varying spectrum of non stationary seismic records. The model is formulated as:

$$S_p(f, t) = S_{k-T}(f, t, Z(t))g^2(t, C), \quad (23)$$

where $g(t, C)$ is expressed in Eq. (21), $S_{k-T}(f, t, Z(t))$ is given by:

$$S_{k-T}(f, t, Z(t)) = \frac{f_g^4(t) + 4\xi_g^2(t)f_g^2(t)f^2}{(f_g^2(t) - f^2)^2 + 4\xi_g^2(t)f_g^2(t)f^2} \cdot \frac{f^4}{(f_f^2(t) - f^2)^2 + 4\xi_f^2(t)f_f^2(t)f^2} S_{g0}(t) \quad (24)$$

in which $Z(t) = (f_g(t), \xi_g(t), S_{g0}(t))$ is a parameter function set of $S_{k-T}(f, t)$, $\xi_f(t) = \xi_g(t)$, and $f_f(t) = 0.15f_g(t)$.

In Eq. (23), the parameter sets C and $Z(t)$ are estimated using the Matlab function `lsqcurvefit` ($\cdot$) for nonlinear curve fitting. The results are shown in Tables 2 and 3, respectively.

Once parameter sets C and $Z(t)$ have been determined, the time-varying modified Kanai-Tajimi stochastic model of the seismic records can be obtained from Eq. (23). Figures 6a–c show the simulated time-varying spectrum, time marginal function, and frequency marginal function. The error values with regard to the time-varying modified Kanai-Tajimi model of the seismic records for the three site classes are calculated from Eq. (22) (Table 3). The error values are compared in Tables 2 and 3 and show that the simulation of the time-varying spectrum using the time-varying modified Kanai-Tajimi model has a higher accuracy than that using the uniform modified Kanai-Tajimi model.

TABLE 3 Set of three parameter functions of three venues

Site classification	Class 1	Class 2	Class 3
$f_g(t)$	$4.10-0.21t+0.0037t^2$	$3.26-0.15t-0.0024t^2$	$2.42-0.10t+0.0014t^2$
$\xi_g(t)$	$1.26-0.029t+0.00036t^2$	$\begin{cases} 1-0.013t & t \leq 15 \\ 0.8 & t > 15 \end{cases}$	$\begin{cases} 1.1-0.020t & t \leq 10 \\ 0.9 & t > 10 \end{cases}$
$S_{g0}(t)$	$\frac{1}{106.16-5.17t+0.088t^2}$	$\frac{1}{91.15-4.07t+0.063t^2}$	$\frac{1}{108.16-4.45t+0.061t^2}$
Error (%)	4.3	2.5	3.8

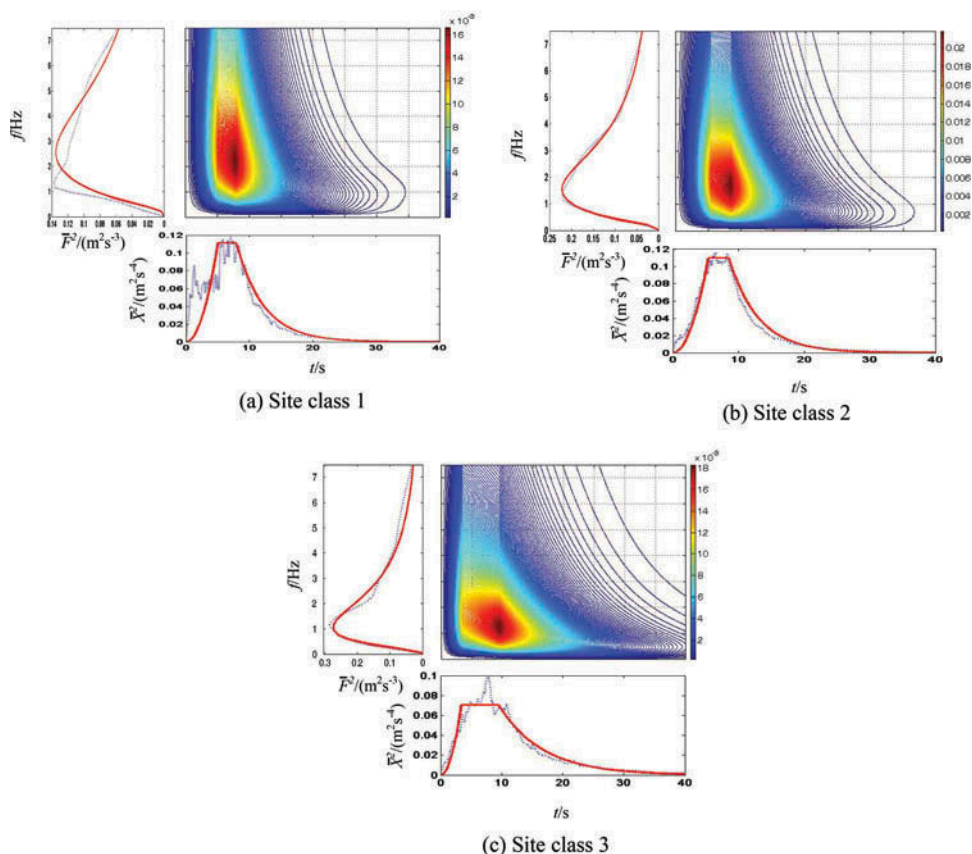


FIGURE 6 Time marginal function, frequency marginal function, and time-varying modified Kanai-Tajimi stochastic model contours of seismic records.

4.3. Determination of the Spectral Intensity Factor

The time-varying spectrum of the seismic records for different intensities and different sites are expressed as the product of the time-varying spectrums of the standardized seismic records ($S_P(f, t)$) and the spectral intensity factor (S_0) as:

$$S(f, t) = S_P(f, t) \cdot S_0, \quad (25)$$

where $S_P(f, t)$ is determined by Eq. (19) or (23).

The time-varying spectrum reflects the energy distribution of the seismic records in the time-frequency space and is given by the following equation:

$$\int_0^{+\infty} \int_0^{t_p} S(t, f) dt df = E \left[\int_0^{t_p} x^2(t) dt \right], \quad (26)$$

where $E \left[\int_0^{t_p} x^2(t) dt \right]$ is the mean energy of the seismic records.

By substituting Eq. (19) or (23) into Eq. (26) and using the condition $\int_0^{+\infty} \int_0^{t_p} S_P(t, f) dt df = 1$, S_0 can be calculated as:

$$S_0 = E \left[\int_0^{t_p} x^2(t) dt \right]. \quad (27)$$

The Chinese Seismic Code of Buildings (CSCB) presents the peak ground acceleration (PGA) of seismic records for different seismic intensities and different sites. The PGAs of the actual records shown in Table 1 are adjusted to be compatible with the CSCB, and Eq. (27) is used to calculate the intensity factor S_0 . The results are listed in Table 4 and show that for the same PGA, the spectral intensity of the seismic records from the soft sites is greater than that from the hard sites. After $S_P(f, t)$ and S_0 have been determined, Eq. (25) can be used as the input to calculate the non stationary seismic response of structures.

5. Generating Artificial Non Stationary Ground Motions

Once the time-varying spectrum of the ground motions has been simulated using the time-varying modified Kanai-Tajimi stochastic model described by Eq. (23), the artificial ground motions can be generated using the stationary Gaussian white noise process filtered through two linear time-invariant filters and multiplied by the amplitude envelope function $g(t)\sqrt{S_{g0}(t)}$. This process can be described by time-varying linear differential equations as follows:

$$\begin{cases} \ddot{x}_f + 2\xi_f(t) \cdot \omega_f(t) \cdot \dot{x}_f + \omega_f^2(t) \cdot x_f = n(t) \\ \ddot{x}_g + 2\xi_g(t) \cdot \omega_g(t) \cdot \dot{x}_g + \omega_g^2(t) \cdot x_g = -\ddot{x}_f \\ a(t) = [\ddot{x}_f + \ddot{x}_g] e(t) \\ e(t) = g(t)\sqrt{S_{g0}(t)} \end{cases}, \quad (28)$$

where x_f and x_g are the responses of the filter, $\omega_g(t) = 2\pi f_g(t)$, $\omega_f(t) = 2\pi f_f(t)$, and $a(t)$ are the synthesized ground motion, $e(t)$ is the time modulating function, and $n(t)$ is the stationary Gaussian white noise process. The parameters are as follows:

$$E[n(t)] = 0, E[n(t_1)n(t_2)] = 2\pi S_0 \cdot \delta(t_1 - t_2), \quad (30)$$

where $E[\cdot]$ is the mathematic expectation and $\delta(\cdot)$ is the Dirac function.

By solving Eq. (28) and using the parametric function set from Table 3, the artificial non stationary ground motion is generated with the time-varying modified Kanai-Tajimi model (described by Eq. (23)). The artificial accelerogram for site class 3 is shown in

TABLE 4 The spectral intensity factor compatible with the Chinese Seismic Code $S_0(m^2 \cdot s^{-3})$

Seismic intensity		VI	VII	VIII	IX
Frequent earthquake	Class 1	0.0277	0.1049	0.4194	1.6778
	Class 2	0.0315	0.1191	0.4763	1.9051
	Class 3	0.0398	0.1503	0.6012	2.4049
Rare earthquake	Class 1	—	4.1430	13.6960	32.9053
	Class 2	—	4.7045	15.5625	37.3462
	Class 3	—	5.9387	19.6329	47.1666

Fig. 7a. The artificial non stationary ground motion with the uniform non stationary model (described by Eq. (19)) is shown in Fig. 7(b). The amplitude of the artificial accelerogram in Fig. 7b is non stationary, but the frequency is stationary, and the amplitude and frequency of the accelerogram shown in Fig. 7a are both non stationary. The non stationary frequency characteristics of this accelerogram may have a large impact on the seismic response of nonlinear structures.

To verify the reliability of the method described above, 100 non stationary ground motions for site class 3 are generated using Fig. 6c as the target time-varying spectrum. The average time-varying spectrum of these artificial ground motions is calculated and is shown in Fig. 8. A comparison of Fig. 8 and Fig. 6c shows a good agreement between the average time-varying spectrum of the artificial ground motions and the target time-varying spectrum.

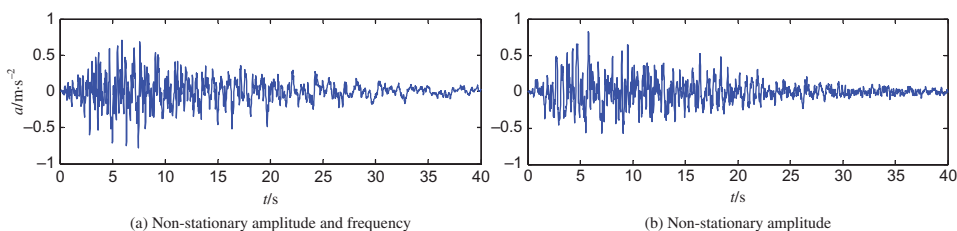


FIGURE 7 Time history of artificially generated ground motion.

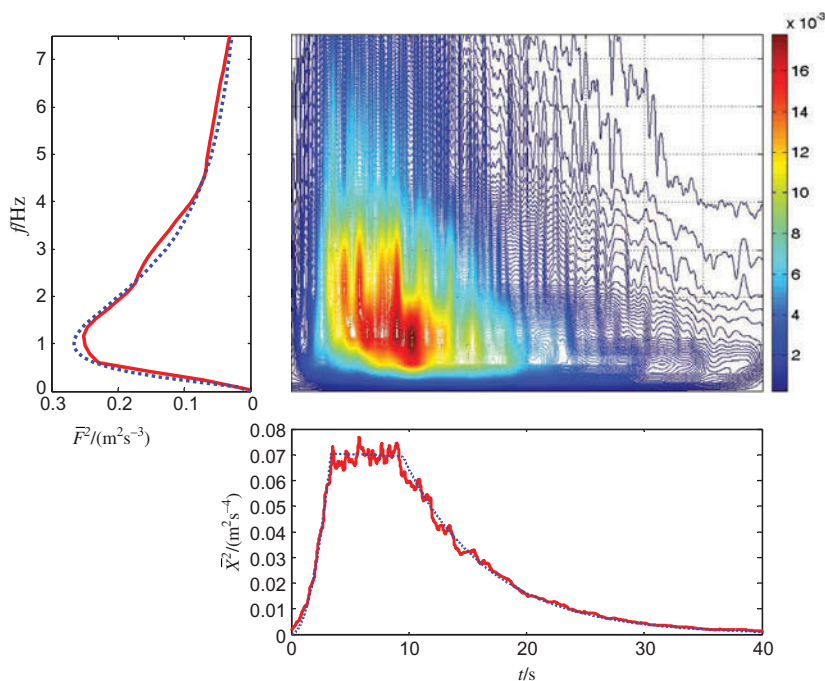


FIGURE 8 Average time-varying spectrum of artificial ground motions (dotted line is the target spectrum).

6. Conclusion

This article estimates the time-varying spectrums of strong ground motions using the S-transform, which is a modern signal analysis technique. The uniform modulated non stationary stochastic model and the time-varying modified Kanai-Tajimi stochastic model of seismic records have been established using nonlinear function-fitting technology. The main conclusions of this article are as follows

1. An algorithm to compute the S-transform is introduced by defining the time-varying amplitude spectrum and the time-varying energy spectrum, and the time-frequency characteristics of the Kobe seismic record are analyzed with different window functions. The results indicate that the S-transform with a complex window is the most effective method of estimating the time-varying spectrum of seismic records when the parameters of the time windows are selected properly.
2. Statistical analyses of the time-varying spectrums of seismic records from different sites show that the seismic records are non stationary in both amplitude and frequency, and their instantaneous center frequencies and instantaneous bandwidths decrease with time. The seismic records contain more high-frequency components in the initial stage and more low-frequency components in the terminal stage.
3. The average time-varying spectrums of the seismic records are simulated using the uniform non stationary stochastic model and the time-varying modified Kanai-Tajimi non stationary stochastic model. Empirical formulas for the time-varying spectrums of the seismic records from three classes of soil sites are established using a nonlinear function-fitting method and can be used as inputs for random seismic response analysis of structures.
4. We proposed a new method to generate artificial non stationary ground motions by solving the time-varying linear differential equations. The average time-varying spectrum of the artificial ground motions agrees well with the target time-varying spectrum, which validates the proposed method.

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